Factors affecting military vehicle trafficability, Pachmarhi, Madhya Pradesh

Assignment section 7 report. Meteorological values are computed from **pachmarhi_daily_2015_2025.csv** — 4,018 daily records, 1 Jan 2015 – 31 Dec 2025, 22.4675 N 78.4336 E, 1,067 m AMSL. **Rainfall is observed** — IMD 0.25° rain-gauge grid for 2015–2024, with ERA5 filling 2025 where the IMD archive has not yet caught up. Temperature, humidity, soil moisture, wind, pressure and cloud are ERA5-Land / ERA5 reanalysis; the IMD data validates ERA5 rainfall at monthly r = 0.939. Non-meteorological factors are catalogued with the source that would supply them.

How to read this. Trafficability is one inequality — **the shear strength the ground can supply against the stress the vehicle demands** — plus two failure modes that bypass it entirely: the route becomes unusable (washout, blocked defile, load-class limit), or the ground is geometrically impassable (boulders, gullies, stems) whatever its strength. Weather enters almost entirely through the *strength* side, by way of soil water. Every factor below is tagged with where the data comes from.

IMD GAUGE observed — IMD rain-gauge grid · ERA5 reanalysis — physical model, observation-constrained · DERIVED computed from this dataset · EXTERNAL another dataset, named · RECCE field survey or imagery · VEHICLE from the vehicle's specification

A Weather — in this dataset

Soil moisture, 0–7 cm ERA5 — the strength of the layer the tyre acts on. *42.0 % v/v monsoon mean vs 26.1 % outside it; 79 % of monsoon days above 40 %.*

Soil moisture, 7–28 cm ERA5 — the layer that carries axle load; tracks the surface at r = 0.96, so wetting is not skin-deep.

Daily rainfall IMD GAUGE — wets the soil and drives runoff. The one variable here resting on real measurements rather than model physics. *1,595 mm/yr mean, 92.3 % of it in Jun–Sep; wettest day 205 mm.*

Relative humidity ERA5 — controls the drying rate, and is the best proxy for soil state where soil data is absent (r = 0.88). *80.4 % monsoon vs 49.5 %.*

Cloud cover ERA5 — suppresses evaporative drying and cuts illumination. *82 % monsoon mean.*

Air temperature (max / min / mean) ERA5 — drying energy, crew endurance, engine and transmission cooling. *Tmax ≥ 35 °C on ~30 d/yr; Tmin ≤ 8 °C on 14 d/yr.*

Wind speed and direction ERA5 — evaporation, dust transport, rotary-wing limits. *2.05 m/s mean; never above 10 m/s in eleven years.*

Atmospheric pressure ERA5 — no direct effect; a synoptic indicator (r = −0.46 with rainfall) and a density-altitude input for aviation.

B Weather-derived indices

Antecedent rainfall, 3-day and 7-day DERIVED — the operative rain quantity, since soil integrates over days. *Maxima 328 mm and 666 mm.*

Drying-window length DERIVED — consecutive dry days available for recovery. *29 wet spells of ≥ 7 days; longest 25 days.*

Heavy-fall frequency DERIVED — washout and flash-flow risk. *3.6 days/yr above IMD's 64.5 mm heavy threshold.*

Rainfall intensity (sub-daily) DERIVED — runoff and erosion scale with intensity, not daily total; recoverable from the hourly source data.

Evapotranspiration / drying rate DERIVED — how fast strength returns after rain; from temperature, RH, wind and radiation.

Fog and low-visibility days DERIVED — proxied by RH > 90 %. *36.8 days/yr, 35 of them in the monsoon.*

C Soil and substrate

Soil texture and USCS class EXTERNAL — decides how much strength is lost per unit of water; the single most important static property. NBSS&LUP soil maps.

Fines content and plasticity (LL, PI) RECCE — high-plasticity clays lose strength catastrophically near the plastic limit.

Vertisol / black-cotton presence EXTERNAL — the swelling clays of the surrounding Narmada valley are the classic no-go soil when wet.

Cone index and rating cone index RECCE — the direct field measurement of bearing strength; RCI accounts for remoulding under traffic.

CBR / shear parameters (c, φ) RECCE — laboratory equivalents used for road and hardstanding design.

Bulk density and compaction state RECCE — a compacted track is far stronger than the same soil freshly disturbed.

Depth to bedrock / soil thickness EXTERNAL — thin soil over Gondwana sandstone behaves as rock; deep soil can bog.

Laterite and murram caps EXTERNAL — locally excellent running surfaces that stay firm wet; worth mapping as preferred routes.

Organic / peaty horizons RECCE — low strength, high compressibility, slow to drain.

D Ground surface and obstacles

Rock outcrop and boulder fields RECCE — a geometric stop: obstacle height against ground clearance, independent of soil strength.

Boulder size and spacing RECCE — spacing decides whether a vehicle threads through or must override.

Loose dry sand and deep dust RECCE — the dry-season failure mode: high motion resistance when *dry*, the mirror image of mud.

Gravel, scree and talus on slopes RECCE — low effective friction; failure is by sliding rather than sinking.

Surface crust RECCE — a dry crust over wet soil gives a firm first pass and a trapped second one.

Leaf litter and duff layer RECCE — slippery when wet, and it hides what is underneath.

Existing ruts and churned ground RECCE — remoulded soil is weaker than undisturbed soil at the same moisture.

Micro-relief and roughness EXTERNAL — termite mounds, tussocks, erosion rills; sets ride speed. From high-resolution DEM or lidar.

E Vegetation and roots

Stem diameter and spacing EXTERNAL — the standard mobility test for forest: override, thread through, or stop. Sal and teak here.

Root mat strength RECCE — a genuine *positive*: a dense root network reinforces weak soil and can carry a vehicle that bare soil would not.

Root plate and buttress obstacles RECCE — surface roots and stumps act as fixed steps and belly hazards.

Canopy closure EXTERNAL — shaded ground dries far more slowly, so it stays weak longer after rain; also blocks air observation and GPS.

Bamboo and dense understorey RECCE — thickets are effectively impassable to wheels regardless of ground strength.

Grass and ground cover RECCE — protects against erosion and rutting; tall grass conceals obstacles.

Deadfall and slash RECCE — episodic blockage after storms; needs pioneer effort to clear.

F Terrain geometry

Longitudinal slope EXTERNAL — traction demand rises as available shear falls; the escarpment approaches to the plateau are the constraint here. DEM (Cartosat / SRTM).

Side slope EXTERNAL — rollover risk and downhill slide; often more limiting than gradient.

Slope length and shape EXTERNAL — a short pitch can be rushed; a long one cannot.

Gullies, ravines and vertical steps EXTERNAL — checked against the vehicle's ditch-crossing and step-climbing limits.

Aspect EXTERNAL — south-facing slopes dry first; a real route-selection lever after rain.

Defiles and choke points RECCE — width limits and tactical exposure, independent of ground.

G Water and drainage

Water table depth EXTERNAL — a shallow water table keeps soil near saturation regardless of recent rain. CGWB well data.

Drainage density and waterlogging EXTERNAL — hollows and flats stay wet for weeks after the same rain that drains off a slope.

Stream and nala crossings EXTERNAL — ford depth, velocity and bed material; the Denwa/Sonbhadra system. India-WRIS gauges.

Flash-flow response time DERIVED — small steep catchments rise and fall within hours of intense rain.

Floodplain and reservoir backwater EXTERNAL — Tawa backwater and Narmada floodplain inundation.

Springs and seepage lines RECCE — locally saturated ground on otherwise dry hillsides.

H The vehicle

Ground pressure / vehicle cone index VEHICLE — the demand side of the inequality: the RCI a vehicle needs for one pass, and for fifty.

Gross weight and payload VEHICLE — a loaded truck and an empty one are different vehicles on the same ground.

Wheeled vs tracked VEHICLE — tracks spread load over a far larger area and tolerate a much wider moisture band.

Tyre size, section width and ply VEHICLE — contact patch area is the cheapest way to cut ground pressure.

Tyre inflation pressure and CTIS VEHICLE — lowering pressure lengthens the contact patch; a central inflation system can change no-go to go in minutes.

Tread pattern and wear VEHICLE — self-cleaning lugs matter in clay; a worn cross-country tyre behaves like a road tyre.

Drive configuration and diff locks VEHICLE — 4×4 / 6×6 / 8×8, and whether torque actually reaches the wheel with grip.

Axle load distribution VEHICLE — the heaviest axle fails first; a rear-biased load digs in on climbs.

Ground clearance and belly shape VEHICLE — decides bellying-out on ruts and mounds.

Approach, departure and ramp angles VEHICLE — geometric limits at ditches, banks and causeways.

Fording depth VEHICLE — with and without preparation; sets which crossings are usable.

Power-to-weight and gearing VEHICLE — sustained gradient capability, especially at reduced power in heat.

Self-recovery equipment VEHICLE — a winch converts many immobilisations into delays rather than losses.

I Operating pattern

Number of passes RECCE — the most underrated factor: each pass remoulds and weakens the soil, so a route good for one vehicle may fail the fiftieth.

Convoy size and spacing RECCE — spacing lets ruts recover and keeps one bogged vehicle from blocking the rest.

Speed and momentum RECCE — momentum carries a vehicle across short soft patches, at the cost of deeper rutting and less control.

Driver skill and route discipline RECCE — line choice and gear selection routinely outweigh a modest difference in ground strength.

Day vs night movement DERIVED — interacts directly with the fog and cloud statistics above.

Recovery and engineer support RECCE — matting, corduroy and dozer effort convert no-go into slow-go.

J Routes and infrastructure

Road class and surfacing EXTERNAL — metalled, gravel or earth track; the difference dominates in the monsoon.

Bridge and culvert load class EXTERNAL — a hard limit that no ground condition can offset.

Causeways and low-water crossings EXTERNAL — overtopped for hours after heavy falls; the usual monsoon cut here.

Carriageway width and hairpin radius RECCE — long vehicles fail on the escarpment hairpins before they fail on grade.

Cut slopes and landslide-prone sections EXTERNAL — blockage risk rises with the same antecedent rainfall that weakens the soil.

Road camber and side drains RECCE — poor drainage softens the verge and narrows the usable width.

Maintenance state RECCE — potholing and edge failure after each monsoon.

K Environmental and regulatory

Protected-area restrictions EXTERNAL — Pachmarhi lies within the Satpura Tiger Reserve and biosphere; in peacetime this can be the binding constraint regardless of ground state.

Forest fire season DERIVED — late dry season; blocks routes and removes the ground cover that resists erosion.

Dust generation DERIVED — convoy dust degrades intervisibility and loads air filters through the long dry season.

Erosion and route degradation DERIVED — repeated movement on wet slopes cuts the gullies that close the route next season.

Wildlife and civil traffic RECCE — speed limits and timing restrictions on shared roads.

★ Ranking of importance, with reasoning

Ranked by how much each factor moves the governing inequality — ground strength against vehicle demand — or bypasses it through route failure. Dynamic factors (weather, moisture) change the answer week to week; static ones (soil type, slope) decide how much the dynamic factors matter at all. **The pairing is the point: soil moisture ranks first only because soil type decides what a given moisture content does to strength.**

#	FACTOR	WHY IT RANKS HERE
	TIER 1 — DECIDES GO / NO-GO · get these right or the rest is noise	
1	**Soil moisture (0–30 cm)**	The dynamic half of the inequality, and the one weather controls. Strength falls non-linearly as saturation approaches, so the transition is a narrow band rather than a gentle slope. Here it swings 26 % → 42 % between seasons, with 79 % of monsoon days above 40 %.
2	**Soil type and plasticity**	Sets the exchange rate between water and strength. The same 40 % moisture is trafficable sandy loam and impassable black-cotton clay — which is why a moisture figure without a soil class is nearly meaningless.
3	**Vehicle ground pressure — tracked vs wheeled, weight, tyres**	The demand side. Nothing else changes the answer as cheaply: a tracked vehicle, or a wheeled one with tyre pressure dropped, crosses ground that stops a loaded 6×6 outright.
4	**Slope, longitudinal and side**	Raises traction demand exactly where available shear is lowest, and adds rollover and slide risk. On the Pachmarhi escarpment it is the binding terrain constraint.
5	**Antecedent rainfall and drying windows**	The mechanism by which weather reaches the soil — not the daily figure, but the 3–7 day accumulation and whether any dry window follows. With 29 spells of ≥ 7 wet days and one of 25, ground here stays degraded for weeks, not hours.
	TIER 2 — STRONG MODIFIERS · turn slow-go into no-go, or rescue it	
6	**Number of passes / convoy size**	Consistently underestimated. Remoulding weakens soil with every vehicle, so trafficability is a property of the *traffic* as much as the ground: pass one is not pass fifty.
7	**Drainage, water table, waterlogging**	Decides how long a soil holds the water that rain delivered. Hollows and shallow water tables stay weak long after slopes have recovered.
8	**Route infrastructure — causeways, culverts, bridge load class**	Bypasses the strength inequality altogether. A hard load limit or an overtopped causeway closes a route while the surrounding ground is perfectly firm.
9	**Surface obstacles — boulders, outcrop, stumps, gullies**	Geometric go/no-go, independent of moisture. Static, so it is surveyed once and reused — but it caps what any weather forecast can promise.
10	**Vegetation — stem spacing, thickets, root mat**	Off-road, stem diameter and spacing decide whether movement is possible at all. Roots cut both ways: a root mat reinforces weak soil, while stumps and root plates are fixed obstacles.
11	**Loose dry sand and dust**	The dry-season mirror of mud: high motion resistance when moisture is at its *lowest*. It breaks the assumption that dry always means trafficable.
12	**Rainfall intensity and flash flow**	Distinct from accumulation. Intensity drives runoff, gully scour and washout — a route-availability failure on a timescale of hours. Here 3.6 days/yr exceed 64.5 mm.
	TIER 3 — SECONDARY OR CONDITIONAL · shape tempo, rarely stop movement	
13	**Visibility — fog, cloud, dust**	Limits speed, night movement and air support rather than mobility itself. Non-trivial here: RH > 90 % on 36.8 days/yr, monsoon cloud cover 82 %.
14	**Driver skill and route discipline**	Ranks low only because it is not a terrain property; in practice line choice regularly outweighs a modest difference in ground strength.
15	**Air temperature**	Affects crew endurance, cooling on sustained climbs and the drying rate — but does not by itself decide passage. ~30 days/yr above 35 °C.
16	**Regulatory restrictions (Satpura Tiger Reserve)**	Placed here for a mobility assessment, but in peacetime it can override everything above it: ground state is irrelevant if entry is not permitted.
17	**Wind and atmospheric pressure**	Effectively negligible for ground movement at this site — 2.05 m/s mean, never above 10 m/s in eleven years. Pressure matters only as a synoptic indicator and for aviation density altitude.

What this dataset can and cannot settle. Of the seventeen ranked factors, this dataset measures or derives five (1, 5, 12, 13, 15) and informs a sixth (11). Ranks 2, 4, 7, 8 and 9 need soil maps, a DEM, groundwater and route data — all obtainable without fieldwork. Ranks 3, 6 and 14 come from the vehicle specification and the movement plan, not from any environmental dataset.

Limitations. Rainfall is observation-based, but it is a **gauge grid, not a Pachmarhi station**: IMD interpolates a station network onto 0.25° cells, and the nearest gauge may be far away. IMD station records for the site itself sit behind a paywall. Every other variable is **reanalysis** — a physical model constrained by assimilated observations — and soil moisture, on which most of this report rests, is the weakest of them: an unassimilated land-surface model state, not a measurement, uncalibrated to local soil texture. The 9 km and 28 km cells average plateau with surrounding lowland, so orographic rainfall maxima are smoothed and extreme daily totals are under-represented. One seam to note: the IMD rain day runs 08:30–08:30 IST against ERA5's 00:00–24:00 IST, so the day definition changes at 1 Jan 2025.